Locomotion

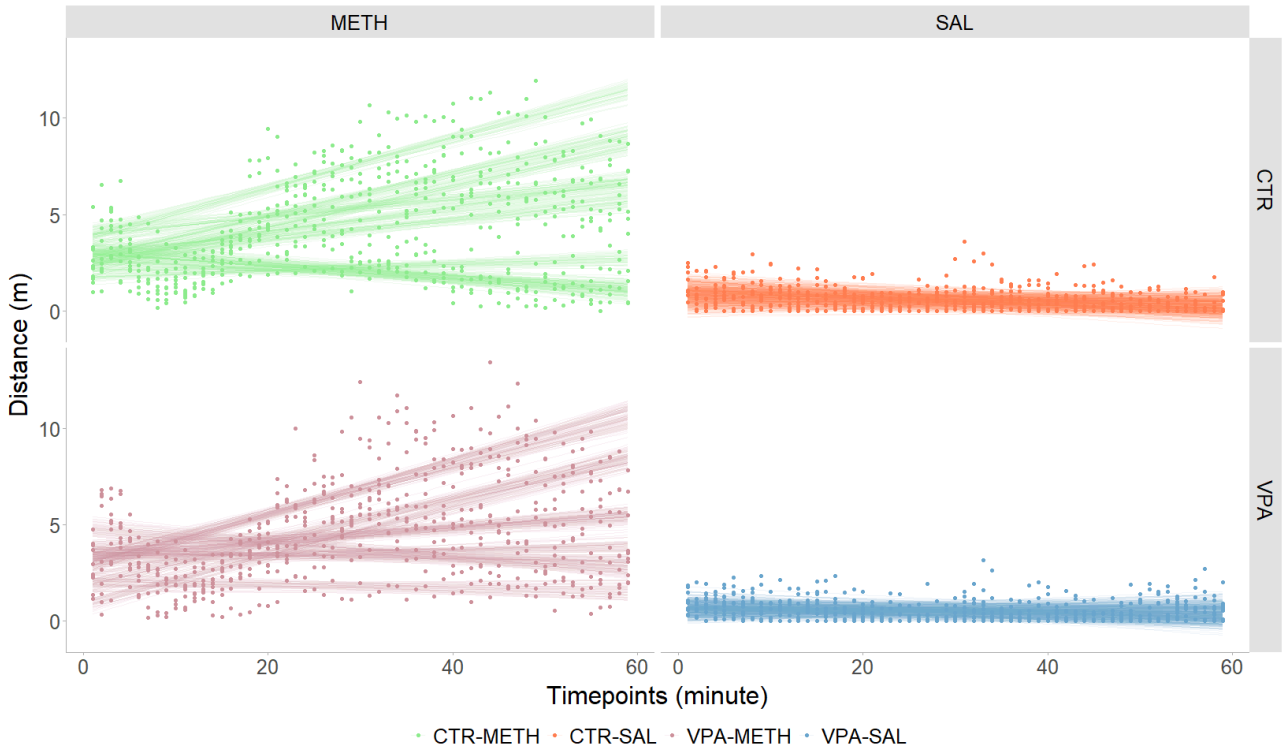
2026-09-30

Table of contents

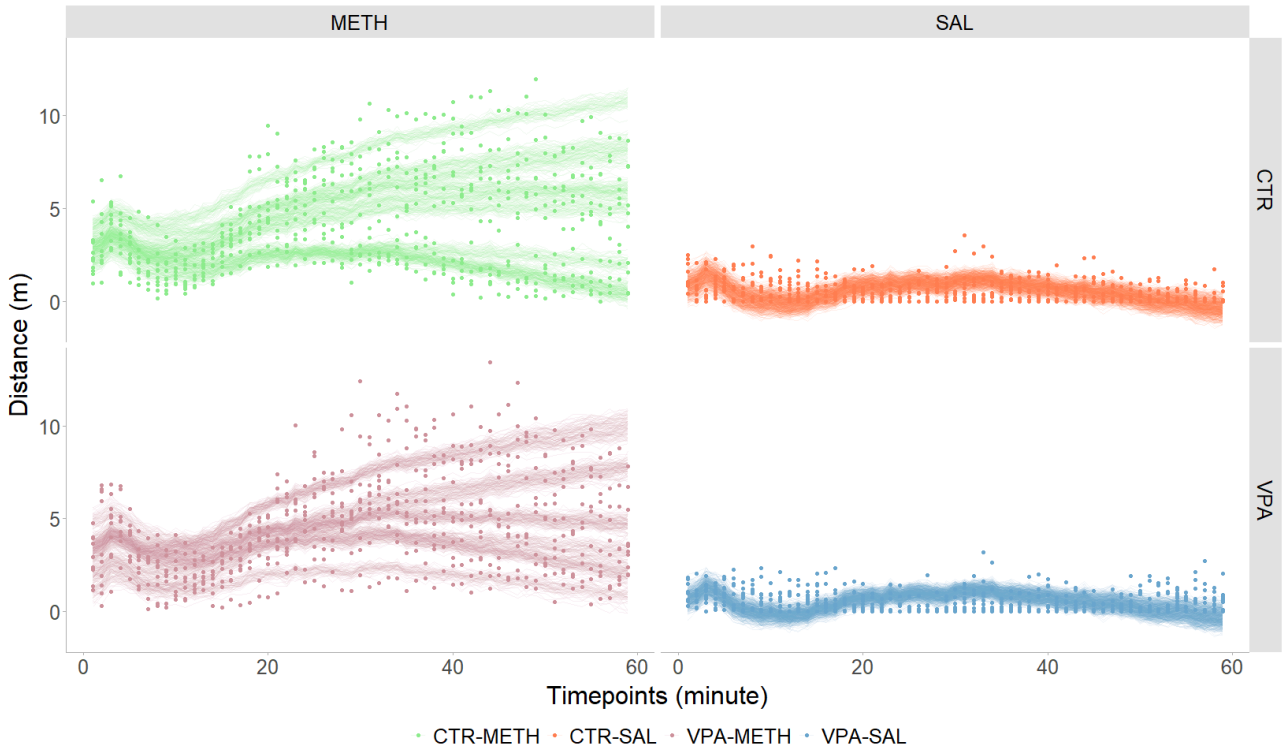
Model options 1
distance_m ~ 1 + minute * group + (1 + minute | animal) + Normal(0, sigma) 1
distance_m ~ 1 + minute * group + (1 + minute | animal) + (1 | minute) + Normal(0, sigma) 2
distance_m ~ 1 + minute * group + (1 + minute | animal) + (1 | minute) + (1 + minute | litter) + Normal(0, sigma) 2
y ~ 1 + gp(time, by = group) + (1 + minute | animal) + (1 + minute | litter) + Normal(0, sigma) .. 3
Most conservative future predictions w/ and w/o RE 3
Contrasts w/ RE (S_Figure 1) 3
Contrasts w/o RE (S_Figure 2) 4
Different random effects contribution 4
No random effects 4
Animal variability 5
Litter variability 5
Time-point variability (the hack) 6
Full variability 6
Most conservative future predictions w/ and w/o RE 7
Contrasts w/ RE (S_Figure 3) 7
Contrasts w/o RE (S_Figure 4) 8

Model options

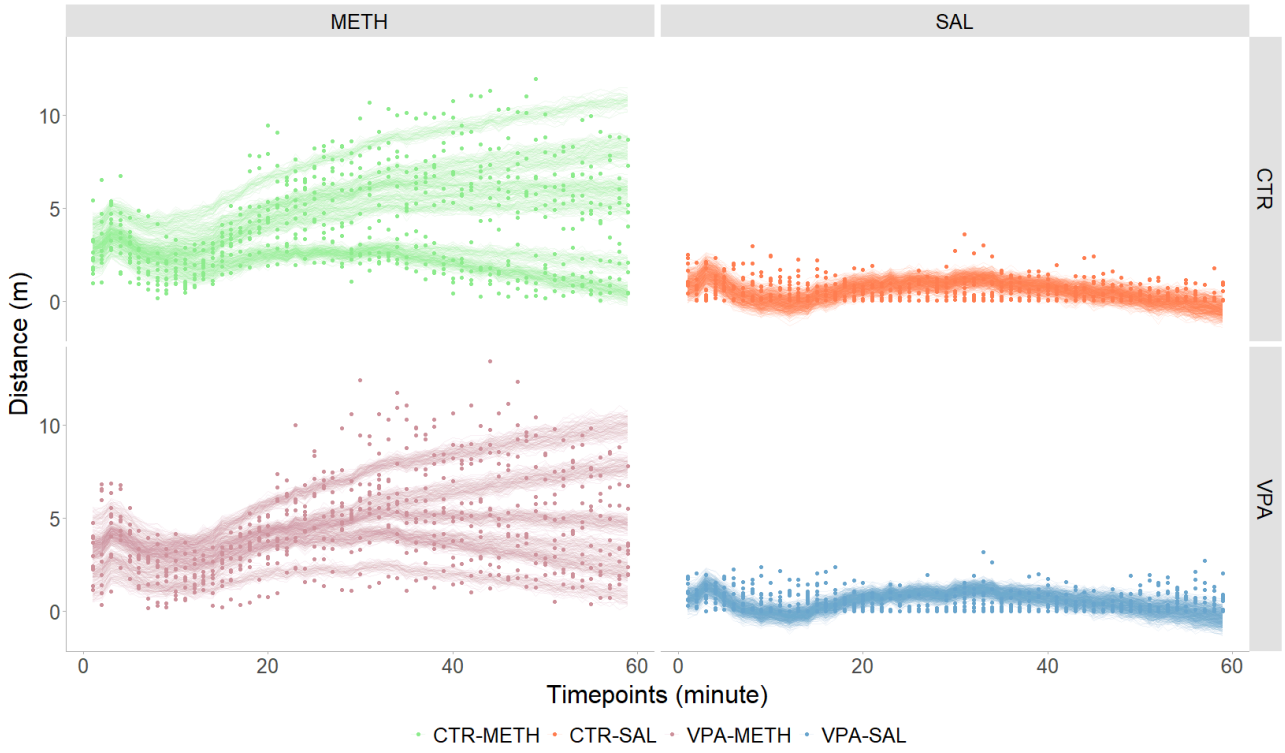
distance_m ~ 1 + minute * group + (1 + minute | animal) + Normal(0, sigma)



distance_m ~ 1 + minute * group + (1 + minute | animal) + (1 | minute) + Normal(0, sigma)

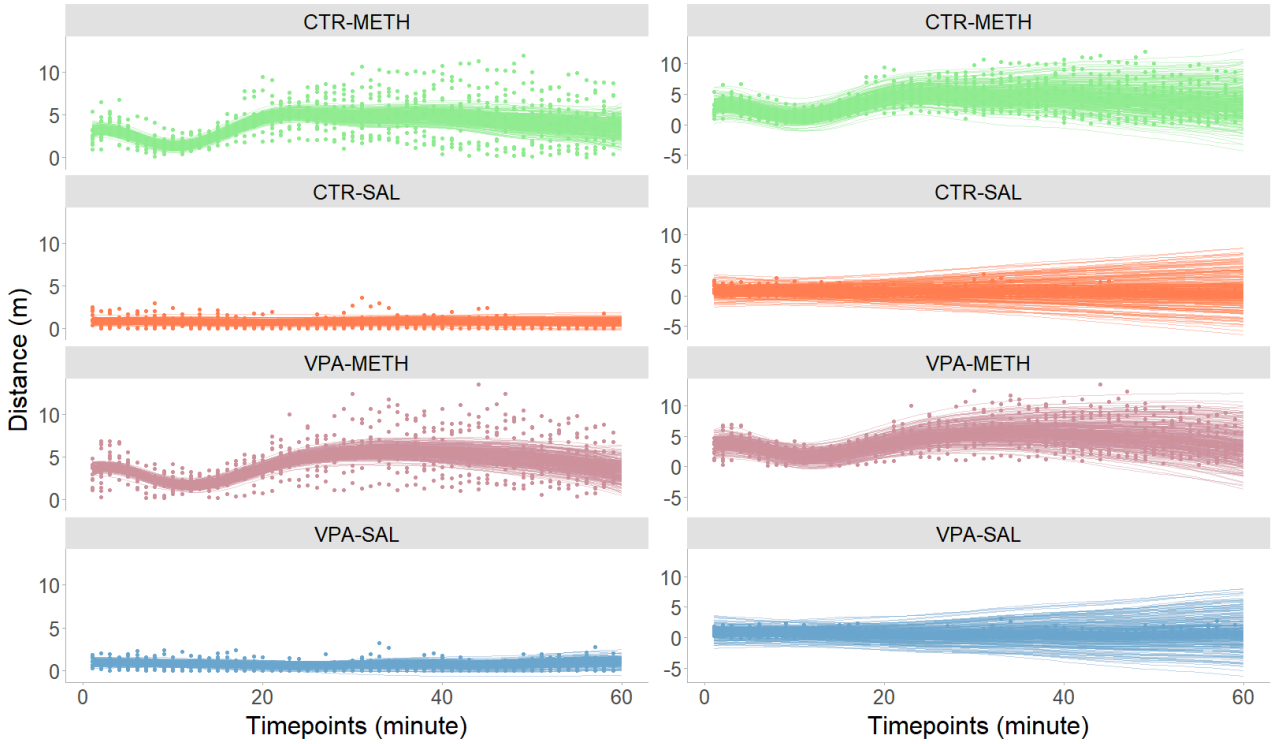


distance_m ~ 1 + minute * group + (1 + minute | animal) + (1 | minute) + (1 + minute | litter) + Normal(0, sigma)

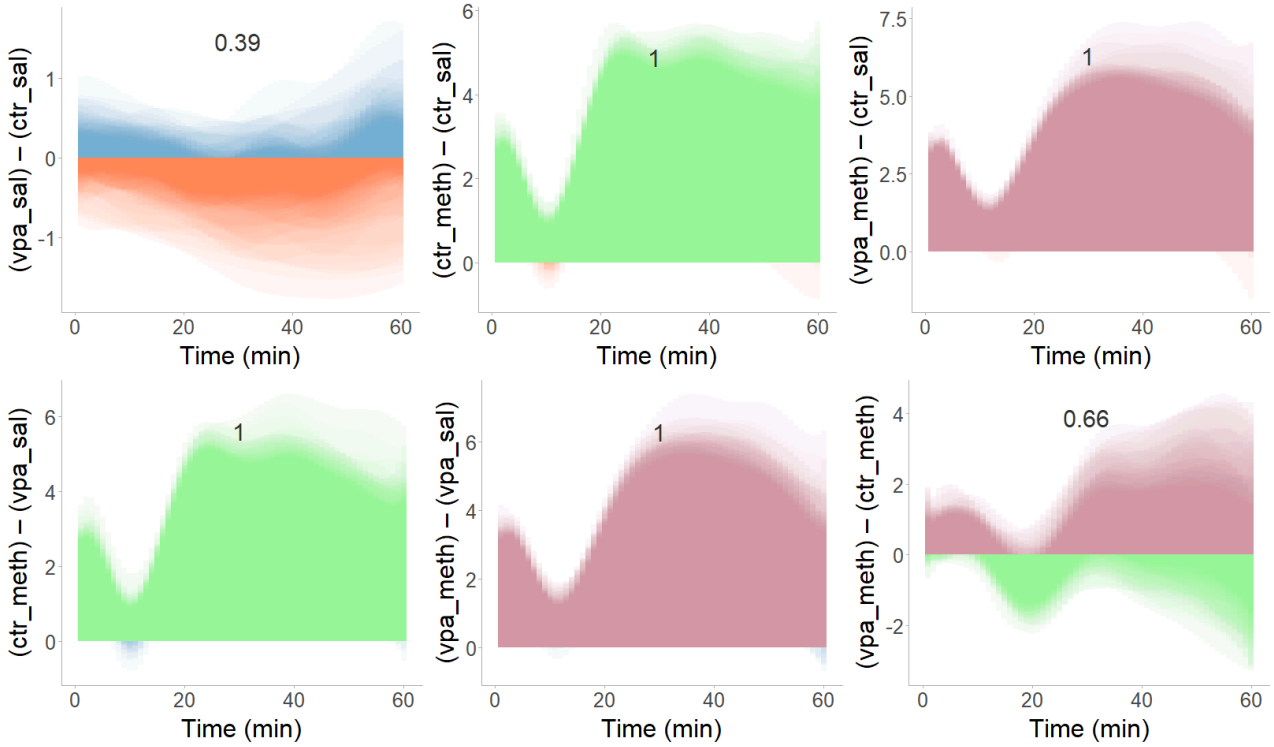


y ~ 1 + gp(time, by = group) + (1 + minute | animal) + (1 + minute | litter) + Normal(0, sigma)

Most conservative future predictions w/ and w/o RE

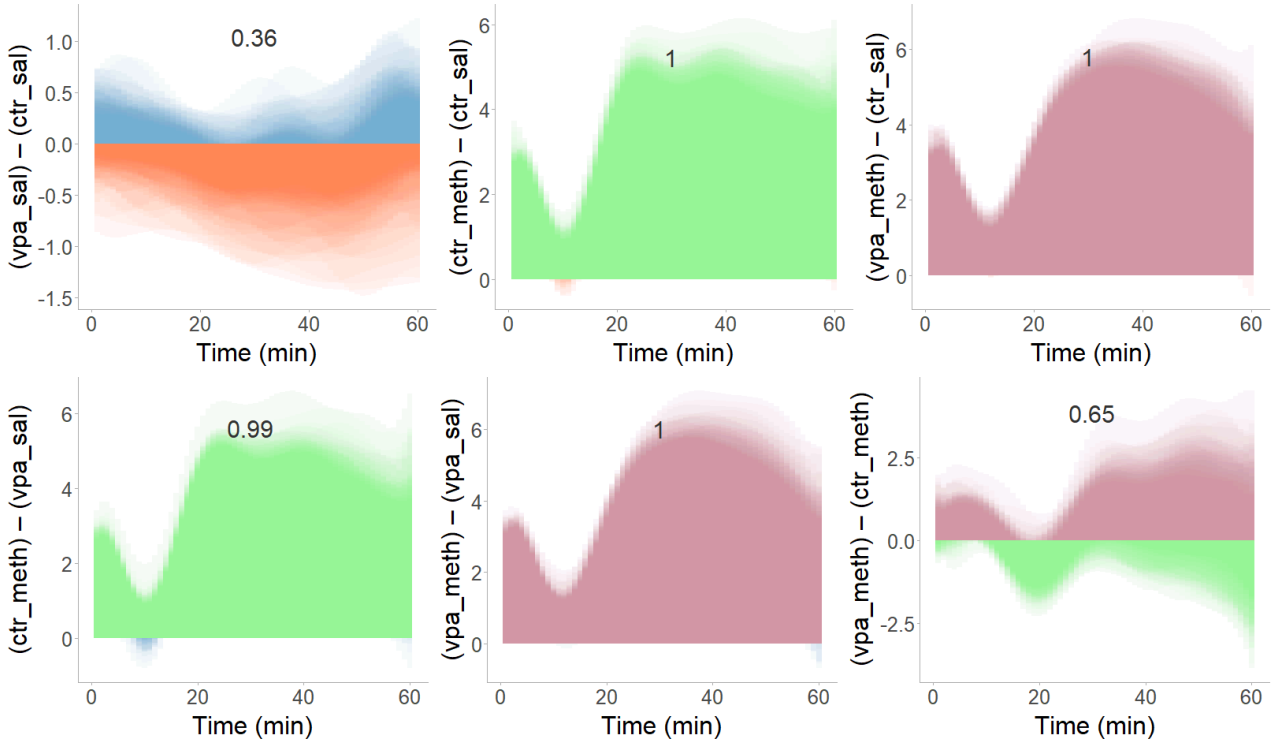


Contrasts w/ RE (S_Figure 1)



S_Figure 1: Contrasts w/ RE

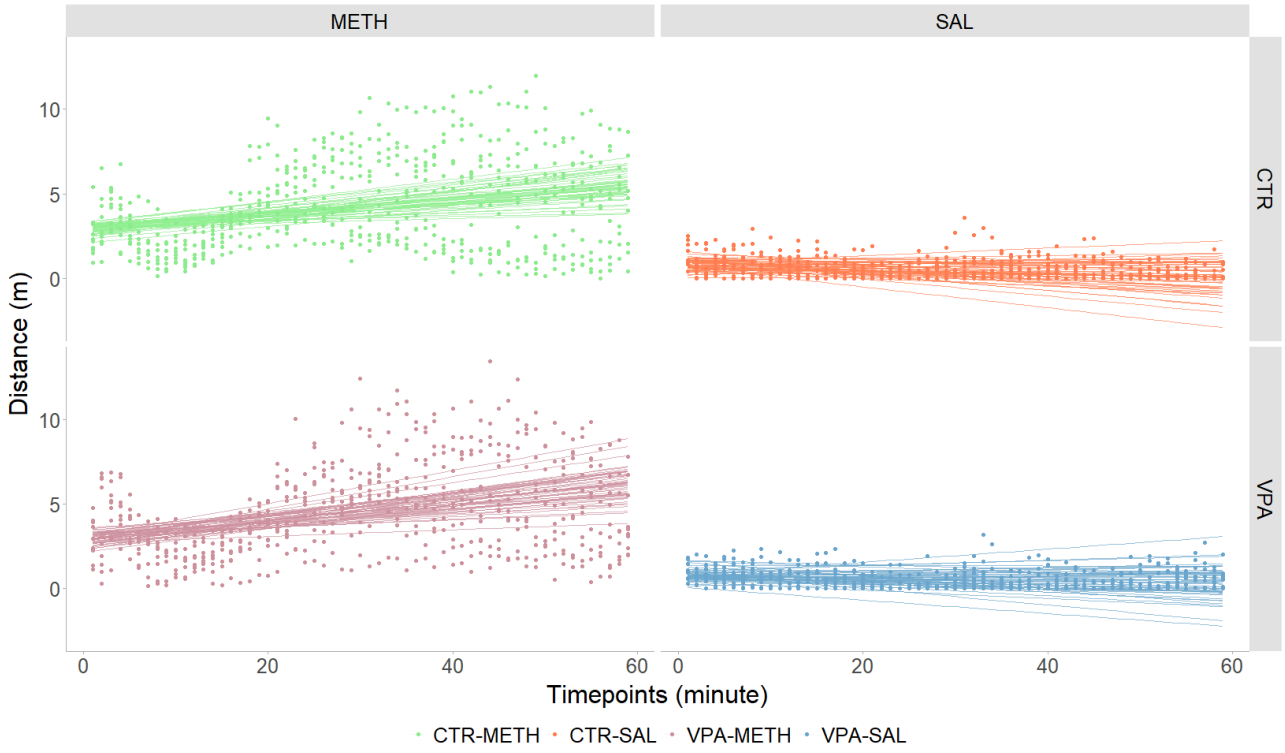
Contrasts w/o RE (S_Figure 2)



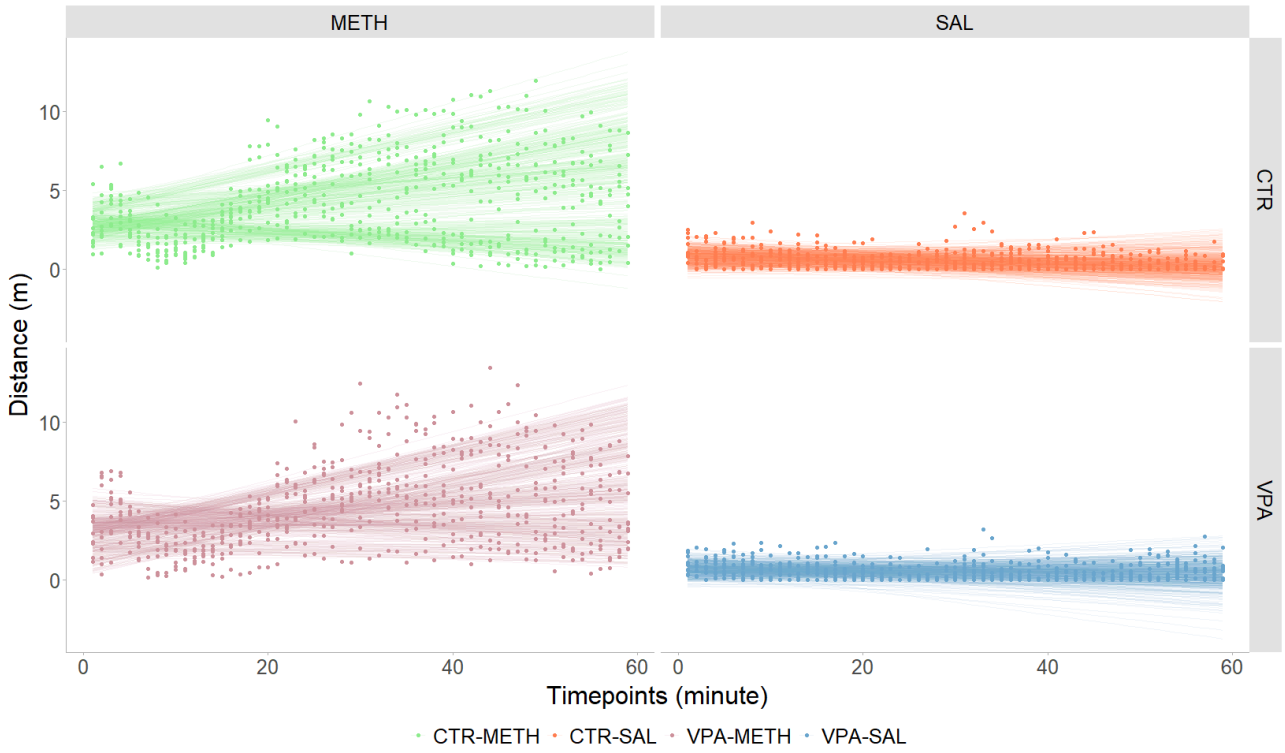
S_Figure 2: Contrasts w/0 RE

Different random effects contribution

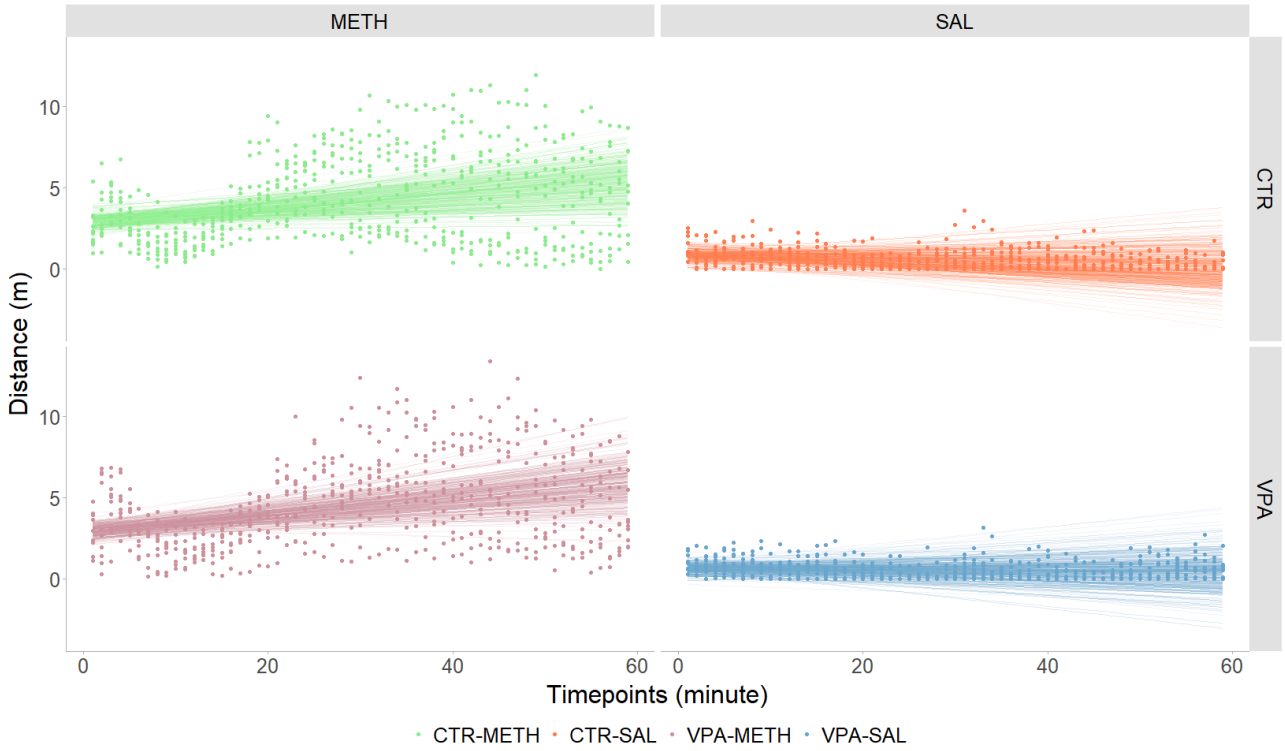
No random effects



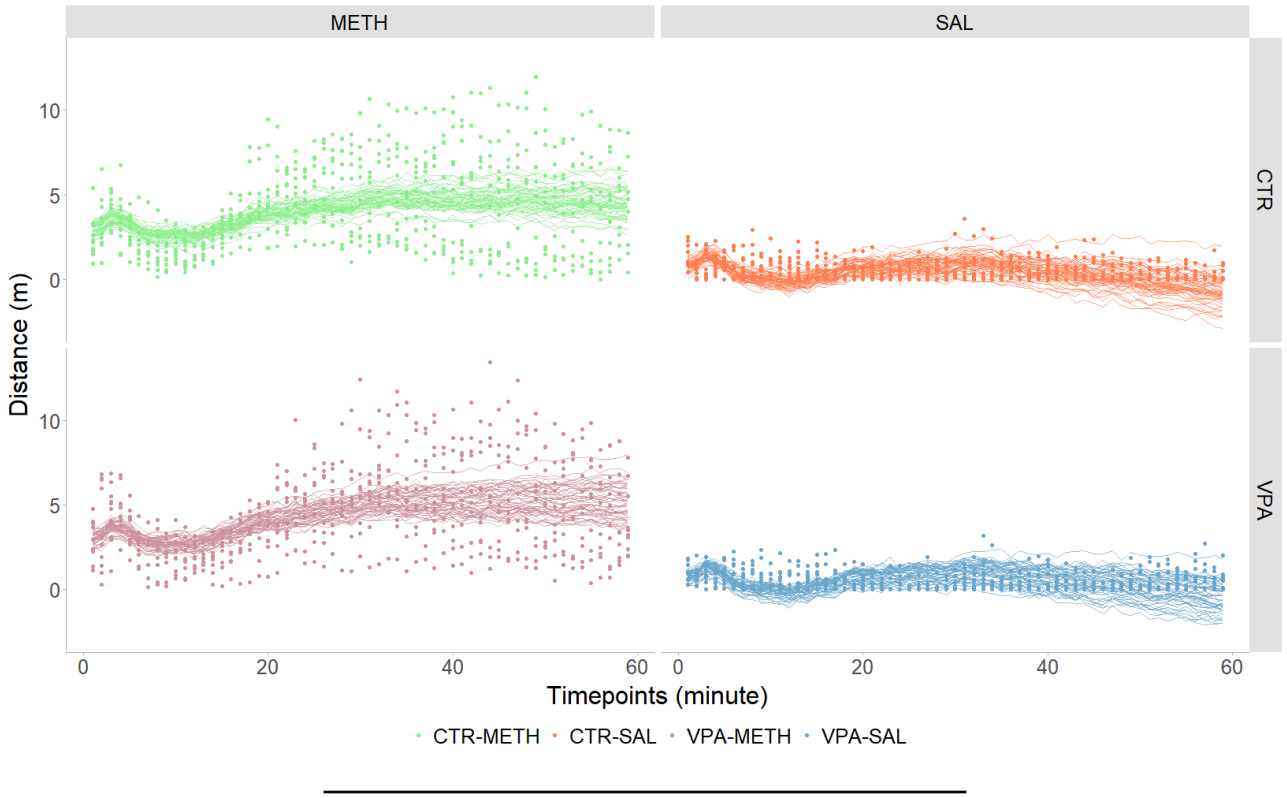
Animal variability



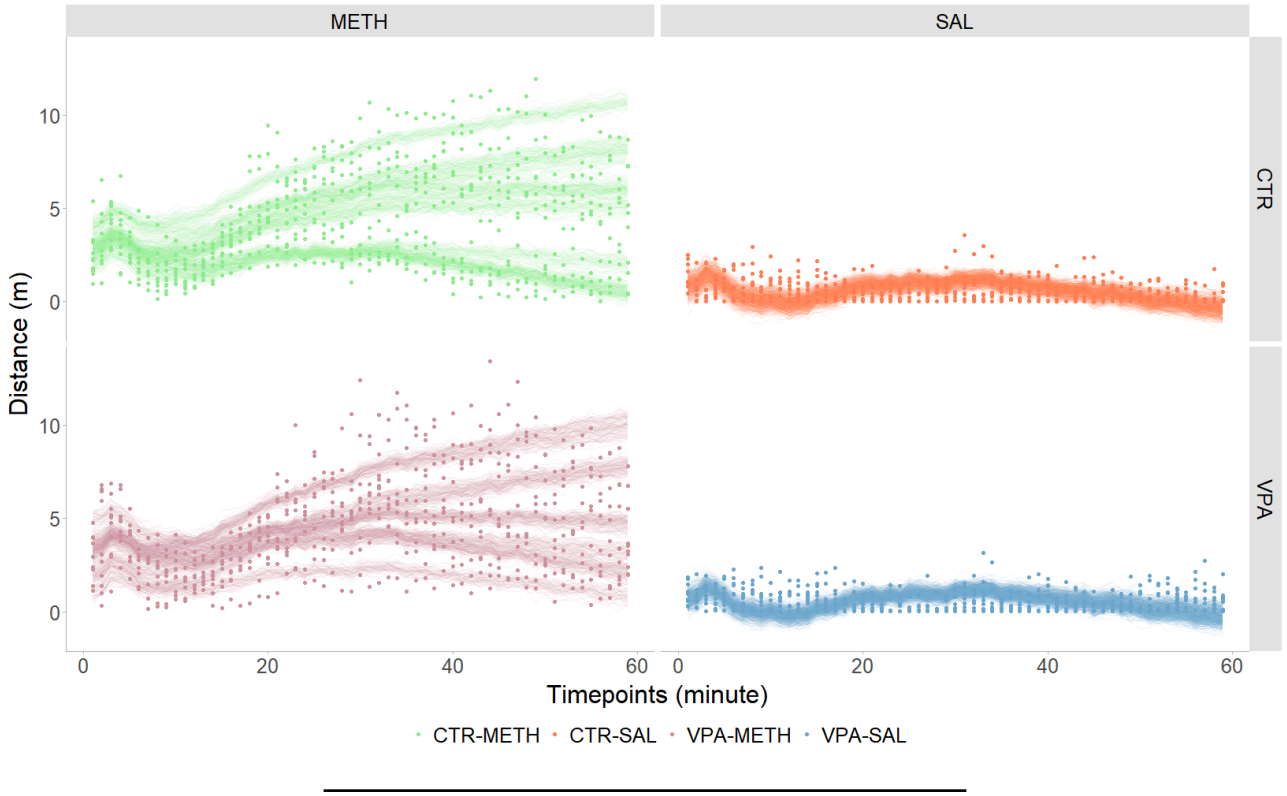
Litter variability



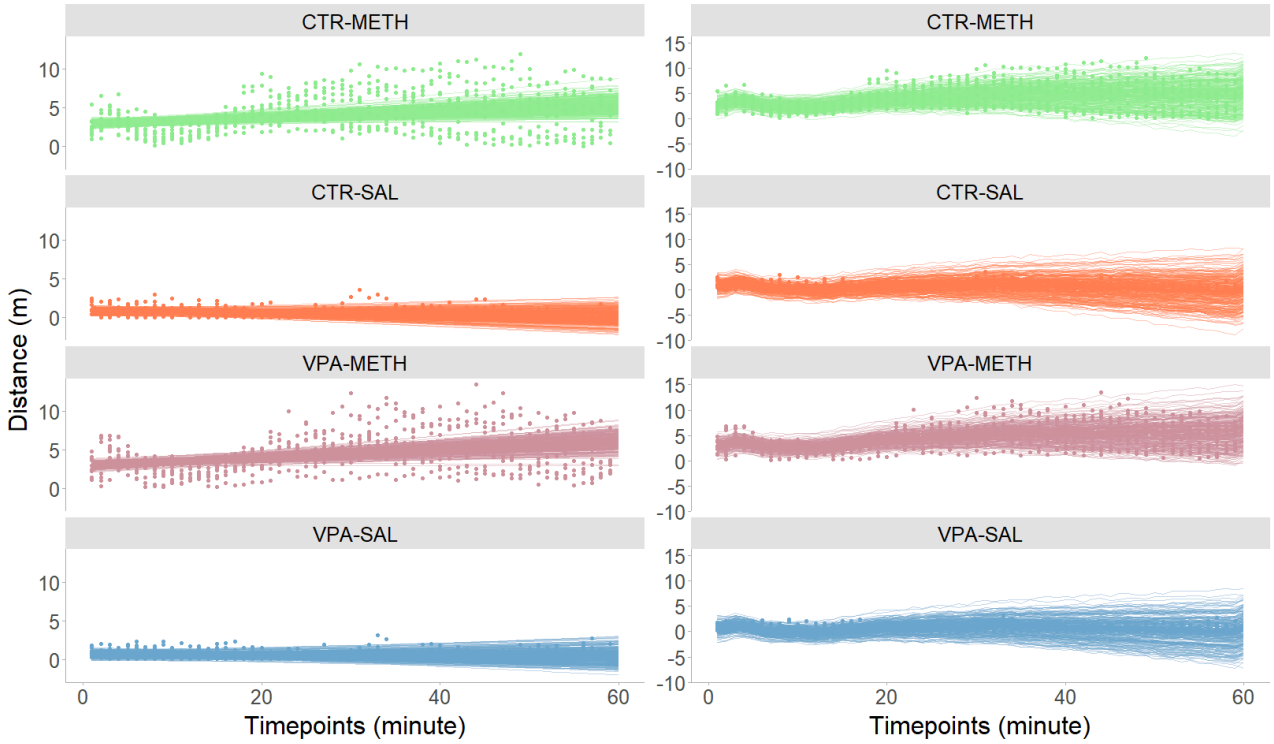
Time-point variability (the hack)



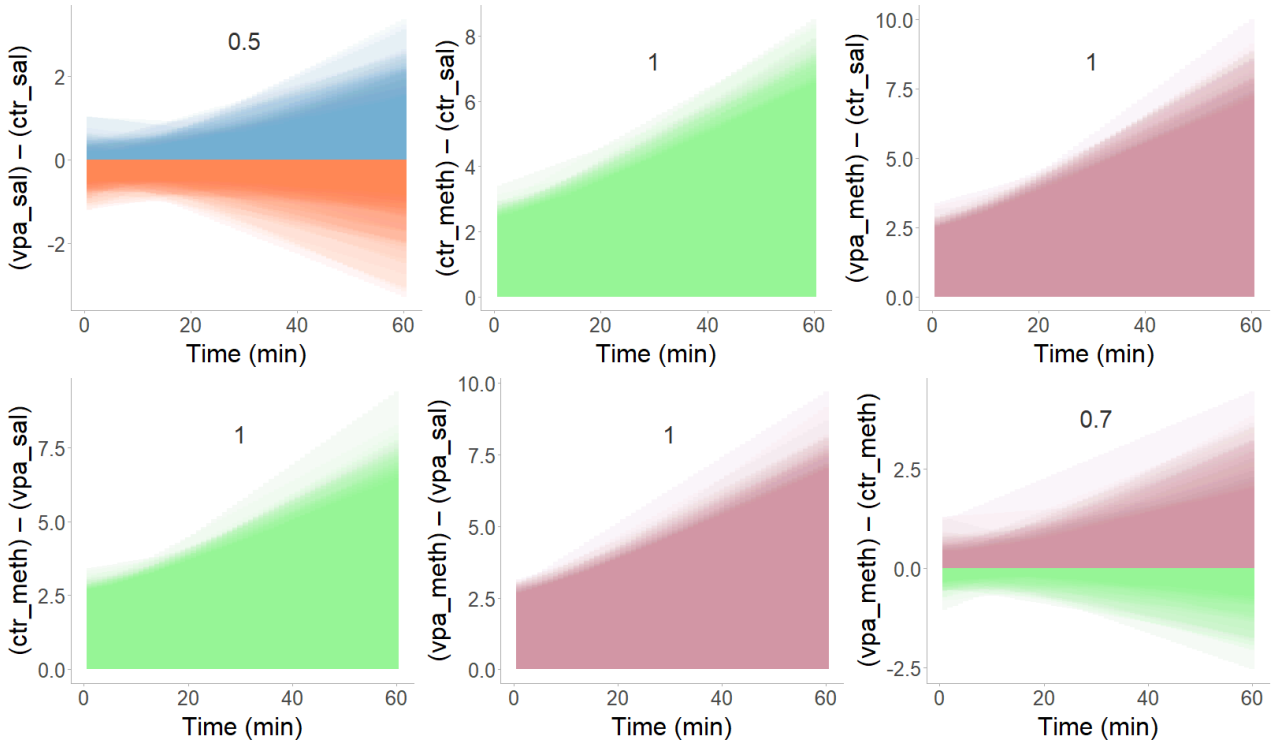
Full variability



Most conservative future predictions w/ and w/o RE

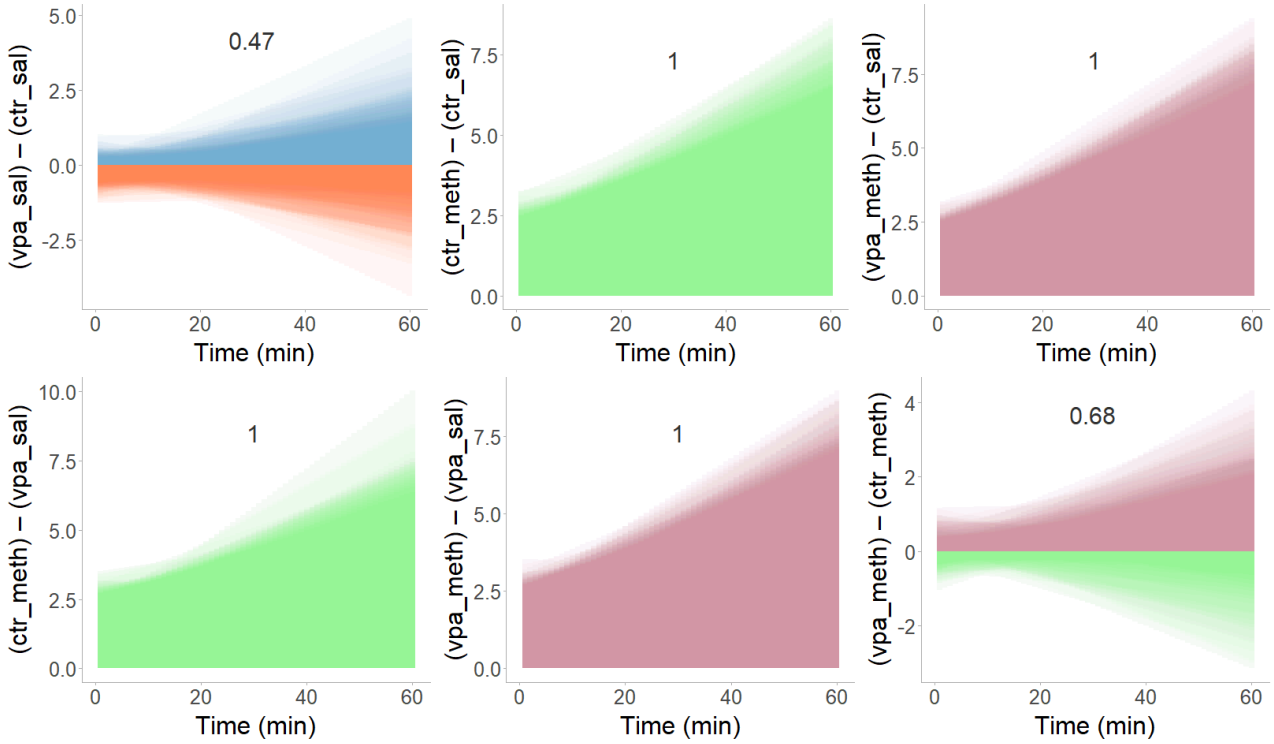


Contrasts w/ RE (S_Figure 3)



S_Figure 3: Contrasts w/ RE

Contrasts w/o RE (S_Figure 4)



S_Figure 4: Contrasts w/o RE